

Algorithm suitability for photonic continuous-variable quantum computers: a comparative resource and noise-robustness study

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Abstract. Photonic continuous-variable (CV) processors are a leading route to scalable quantum computing, yet the fault-tolerance cost of an algorithm depends on its use of controlled unitaries, non-Gaussian gates, and depth. We port four algorithms—control-free phase estimation, randomized product formulas, quantum Kolmogorov–Arnold networks, and Regev factoring—to a common CV model and compare them under photon loss and Gaussian displacement with Gottesman–Kitaev–Preskill (GKP) correction. Control-free phase estimation is the most noise-robust, needing no controlled unitaries and, for quadratic Hamiltonians, no non-Gaussian gates. Correction helps only above the finite-squeezing floor; a noise- versus systematic-limited dichotomy predicts which algorithms benefit.

Photonic continuous-variable (CV) architectures encode information in the quadratures of bosonic modes and support deterministic, room-temperature generation of large cluster states [1, 2, 3, 4, 5], making them a leading candidate for scalable fault-tolerant quantum computing. Fault tolerance is supplied by bosonic codes, most prominently the Gottesman–Kitaev–Preskill (GKP) code [6], whose overhead is set by the achievable squeezing and the number of non-Gaussian resource states consumed [7, 8, 9, 10]. A defining constraint is that Gaussian errors cannot be corrected by Gaussian operations [11], so non-Gaussian resources are mandatory.

Different algorithms place very different demands on this substrate: some require controlled unitaries, which are costly on CV hardware because they need a clean control mode and high non-Gaussianity; some are dominated by non-Gaussian gate count; some by circuit depth. A unified comparison of which algorithm classes are intrinsically robust on CV hardware, and of how much GKP correction improves them, has been lacking. Here we port four representative algorithms to a common CV model, compare them under a shared photon-loss and displacement channel, and identify a finite-squeezing threshold and a noise-versus-systematic-limited dichotomy that together predict robustness and correction gain. We are explicit about which results are exact, which are validated model surrogates, and which are extrapolations.

We map each algorithm onto bosonic modes encoded, where fault tolerance is required, in GKP qubits. Gaussian operations (interferometers, squeezers, phase rotations) are treated as inexpensive; non-Gaussian gates (cubic-phase, GKP magic states) are the costly resource, analogous to the T -count of discrete-variable (DV) circuits.

The four algorithms span the axes that govern CV cost. (i) Control-free phase estimation [12] acquires the time series $f(t) = \langle \psi | e^{iHt} | \psi \rangle$ without controlled unitaries and recovers the spectrum by reference-eigenstate holographic phase retrieval; for quadratic H , e^{iHt} is purely Gaussian. (ii) Randomized product formulas (qDRIFT) [13, 14] approximate e^{-iHt} by importance-sampled bosonic terms with a step count independent of the number of terms. (iii) Quantum Kolmogorov–Arnold networks [15, 16] realize quantum singular value transformation [17, 18, 19], with Chebyshev response $T_r(x) = \cos(r \arccos x)$ that maps to phase-space rotations. (iv) Regev factoring [20, 21]

prepares a discrete-Gaussian GKP comb, applies a mode-wise CV quantum Fourier transform, and runs modular exponentiation on GKP qubits.

Per noise-exposed operation we apply photon loss (transmissivity η) as an exact Fock-space Kraus channel, $E_l = \sqrt{(1-\eta)^l/l!} \eta^{\hat{n}/2} a^l$, and Gaussian displacement (standard deviation σ_d) as an exact unitary kick; the dynamical evolution is simulated exactly in truncated Fock space. GKP correction reduces the residual to

$$\sigma_{\text{corr}}^2 = (1 - p_L) \Delta^2 + p_L \sigma_{\text{phys}}^2, \quad (1)$$

where $\sigma_{\text{phys}}^2 = \sigma_d^2 + (1 - \eta)/2$, Δ (dB) is the finite-squeezing peak width, and p_L is the single-round GKP logical error rate from the standard shift-error model, $p_L = 1 - [\text{erf}(\frac{\sqrt{\pi}/2}{\sqrt{2}\sigma_{\text{eff}}})]^2$ with $\sigma_{\text{eff}}^2 = \sigma_{\text{phys}}^2 + \Delta^2$. We validate Eq. (1) against an explicit Glancy–Knill error-correction Monte Carlo [22]: the analytic p_L reproduces the protocol to within 3×10^{-4} absolute, the post-correction residual equals the floor ($\sigma_{\text{res}}/\Delta = 1.00$), and the surrogate matches the explicit two-channel model to within 0.1%. Stochastic quantities use ten random seeds with 95% confidence intervals; all scripts and data reproduce every figure.

Robustness. Under increasing displacement, control-free phase estimation keeps its error near the clean floor (1.8×10^{-3}) up to $\sigma \approx 0.20$, whereas qDRIFT, the network model, and Regev are already degraded by $\sigma \approx 0.05$. This ordering follows from the absence of controlled unitaries and, for quadratic H , non-Gaussian gates in phase estimation.

Floor crossing. From Eq. (1), $\sigma_{\text{corr}} < \sigma_{\text{phys}}$ if and only if $\sigma_{\text{phys}} > \Delta$, because $\sigma_{\text{corr}}^2 - \sigma_{\text{phys}}^2 = (1 - p_L)(\Delta^2 - \sigma_{\text{phys}}^2)$ and $1 - p_L > 0$. GKP correction therefore reduces error only when the physical noise exceeds the finite-squeezing floor Δ (dB), and is counterproductive below it. Figure 1 maps the improvement factor versus squeezing at a fixed operating point ($\sigma_d = 0.10$, loss 0.15); the floor crossing is evident, and at 18 dB the factors are 1.1 (phase estimation), 1.4 (qDRIFT), 2.5 (network), and 2.7 (Regev).

Noise- versus systematic-limited dichotomy (empirical observation). The error of each algorithm is well described by an envelope $\max(F, c\sqrt{D}\sigma)$, with an algorithm-specific systematic floor F (e.g. finite acquisition window for phase estimation), depth D , per-step residual σ , and an algorithm-dependent constant c fit from the clean-noise sweep. In the noise-limited regime

Table 1. Empirical fits of the dichotomy envelope $\text{err}^2 = F^2 + c_{\text{eff}}^2 \sigma_{\text{phys}}^2$ over the loss sweep ($\sigma_d = 0.10$, loss $\in \{0, 0.02, 0.05, 0.10, 0.20\}$). Units of F and c_{eff} are each algorithm’s natural error unit.

algorithm	F	c_{eff}	R^2
phase estimation	3.0×10^{-3}	3.1×10^{-3}	0.99
qDRIFT	0.47	1.5	0.89
network model	0	0.96	0.95
Regev	21	1.9×10^2	0.97

($c\sqrt{D}\sigma > F$) the GKP gain is $\approx \sigma_{\text{phys}}/\sigma_{\text{corr}} > 1$, realised most strongly by deep, noise-sensitive algorithms (Regev and the network model). In the systematic-limited regime ($F > c\sqrt{D}\sigma$) corrected and uncorrected errors coincide at F and the factor tends to unity. Control-free phase estimation is systematic-limited up to $\sigma \approx 0.2$: it is the most accurate yet shows the smallest improvement factor, reconciling “most robust” with “smallest correction gain”. Least-squares fits of $\text{err}^2 = F^2 + c_{\text{eff}}^2 \sigma_{\text{phys}}^2$ over the five-point loss sweep (Table 1) give $R^2 \geq 0.89$ and locate each algorithm in its regime: F dominates for phase estimation (its systematic FFT floor) and sets the qDRIFT algorithmic floor (finite- N Trotter-like error), while $c_{\text{eff}} \sigma_{\text{phys}}$ dominates for the network model and Regev.

Phase diagram and ensembles. Figure 2 extends the floor crossing across the (σ_d , loss) plane: the helpful region is broad for deep algorithms and narrow for phase estimation. Over ensembles of twenty random Hamiltonians, qDRIFT’s clean channel error is 0.17 ± 0.07 , and control-free phase estimation recovers peak frequencies to median 1.7×10^{-3} with full success provided the holographic reference beam is sufficiently strong—an explicit applicability condition.

Generalization to non-Gaussian Hamiltonians. For $H = \omega \hat{n} + \frac{g}{2}(a^2 + a^{\dagger 2}) + \chi \hat{n}^2 + \kappa x^3$, holographic recovery returns the absolute spectrum with error $\approx 1.3\text{--}1.7 \times 10^{-3}$ that is independent of the non-Gaussian fraction (from 0 to 0.83), while the controlled-unitary count remains zero. The control-free property is thus a feature of the protocol, not of H being quadratic; non-Gaussian H only adds the cost of a Trotterized e^{iHt} .

Discrete-variable baseline. Against a rotated surface code at physical error 3×10^{-3} , reaching logical error 10^{-6} needs distance 17, i.e. 577 physical qubits per logical qubit; a single-round GKP mode at $\sigma_{\text{phys}} = 0.10$ needs ≈ 14 dB, and floors above 10^{-6} for $\sigma_{\text{phys}} \gtrsim 0.2$, requiring concatenation [23, 24]. Structurally, a spectroscopy task (10 readout bits, 8 system qubits, Trotter depth 50) costs 1023 controlled evolutions and a T -count $\sim 2 \times 10^7$ in DV phase estimation, versus zero controlled unitaries and zero non-Gaussian gates

in the control-free CV protocol for quadratic H . The CV advantage is structural—Gaussian operations are free—rather than a threshold effect.

Unit-independent ranking. Mapping each error to a dimensionless infidelity by a monotone soft-saturating transform $I = \text{err}/(\text{err} + s)$ with task-failure scale s preserves the ordering $I_{\text{PE}} = 0.003 < I_{\text{net}} = 0.25 < I_{\text{qDRIFT}} = 0.40 < I_{\text{Regev}} = 0.98$; the ordering is invariant when every s varies over $[0.5, 2] \times$ its nominal value.

Regev squeezing scaling. With $R = e^{\Theta(\sqrt{n})}$ and dual tolerance $\delta = \sqrt{d}/(\sqrt{2}R)$, the comb readout constraint $\sigma_{\text{cv}} = \Delta/\sqrt{\pi} \leq \delta$ gives $\text{dB} \geq 10 \log_{10}(R^2/\pi d) = \Theta(\sqrt{n})$. The unrescaled comb therefore demands squeezing growing as $\sqrt{n}$, prohibitive at cryptographic sizes; we present Regev as a worked correctness check and a cautionary resource case rather than a practical CV factoring route.

The findings cohere into a quantitative selection rule for CV hardware: minimize controlled unitaries, non-Gaussian gate density, and depth. Control-free phase estimation minimizes all three and is the most robust; GKP correction repays its squeezing cost only above the finite-squeezing floor and in proportion to how noise-limited an algorithm is; and CV-native cheapness of state preparation and Fourier transforms does not rescue a depth- and readout-heavy algorithm such as Regev. These criteria are model-level: the dynamical channels are exact, the GKP correction is a surrogate validated to $\leq 0.1\%$ against an explicit error-correction protocol but not a full Fock-space circuit with loss-correcting amplification, and instances are modest in size. The natural experimental target is a GKP-encoded photonic cluster-state processor at 10–15 dB squeezing, where the floor crossing and the phase-estimation-versus-deep-algorithm dichotomy are directly testable; control-free spectroscopy of a small bosonic or Fermi–Hubbard Hamiltonian is the most accessible first demonstration.

The scripts and numerical results that reproduce all figures and tables are available from the authors on reasonable request.

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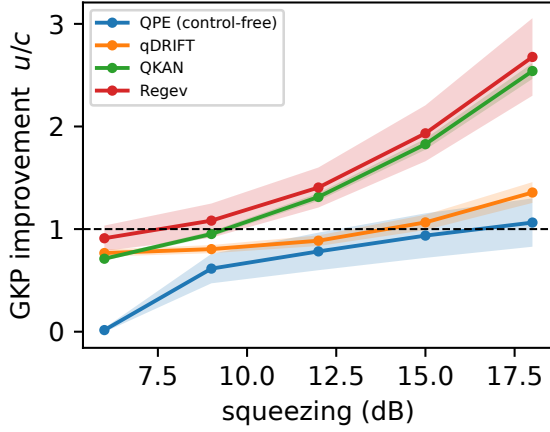


Figure 1. GKP improvement factor versus squeezing at $\sigma_d = 0.10$, loss 0.15; shaded bands are 95% confidence intervals over ten seeds. The crossing of unity marks the finite-squeezing floor.

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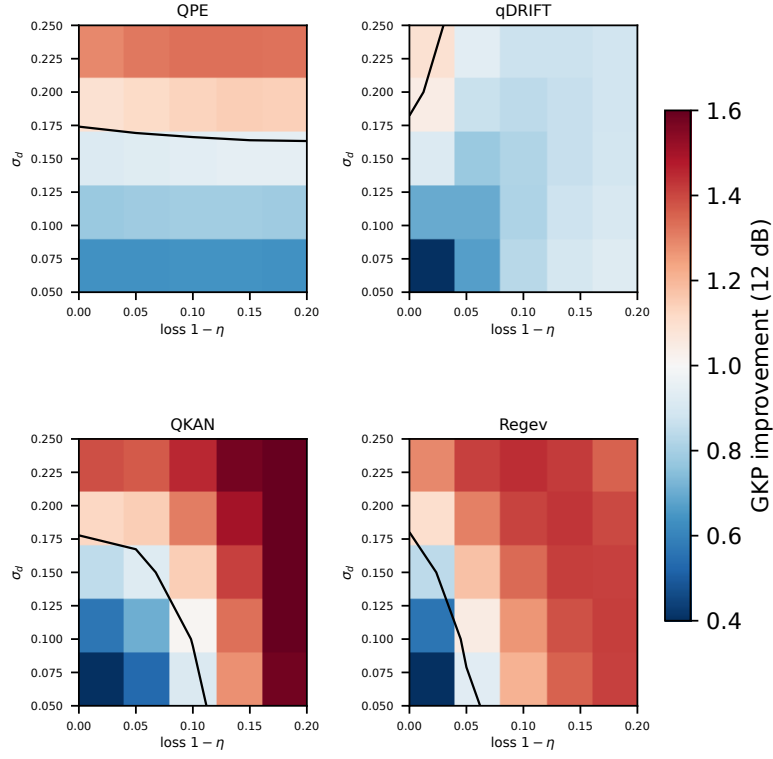


Figure 2. GKP improvement factor (12 dB) over the (σ_d, loss) plane; the black contour marks the floor crossing (factor unity). Blue: correction helps; red: counterproductive.